

Shirin Shahabi

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PROFILE

Machine Learning Engineer and Researcher with 4 years of hands-on experience in the Finance and Tech industries, holding an **MSc in Computer Science** from McMaster University. Proven track record in delivering production-grade AI solutions and expert in **Reinforcement Learning, Model Optimization, Distributed Inference and LLM Evaluation**. Specializing in **Agentic AI security, Inference Optimization, and scalable traceable ML solutions** tailored for high-stakes environments involving risk and compliance.

Cloud expertise in **AWS(Certified)** (Lambda, EC2, SageMaker, Aurora, Airflow, Elasticsearch), and **GCP** (Vertex AI, Kubernetes, VM instances).

EXPERIENCE

Machine Learning Engineer

Inference Labs Inc

May 2024 – Present

Remote, Canada

- Co-engineered and Developed **DSperse**, an **open-source patented software** for distributed, verifiable inference, Improved inference time and optimized memory with fallback fault-tolerant mechanism ([Sourcecode](#)).
- Deployed [verifiable OpenClaw](#), leveraging DSperse computation traces to eliminate prompt injections by design.
- Designed a **SOTA LLM benchmark** to validate reasoning fidelity in prediction market under drift ([Publication](#)).
- Developed distributed, verified inference for an external **In-production, commercial Vision aviation model** using DSperse and JSTProve preserving **Inference fidelity**, served as auditing and compliance tool.
- Collaborated on development of JSTProve; ML toolkit that produces proofs of AI inference by ONNX subgraph extraction and **Quantization**, FFT decomposition, and **graph manipulation(GraphSurgeon/TensorRT)**.
- Trained the industry's first [Verifiable AI Agent](#) integrating proof of inference, Achieved a computational overhead reduction from 1000x to 8x via JIT access, Redis caching.
- Developed safety proxy for **AI browser agents** leveraging DSperse masking for prompt injection detection.
- Co-Authored the **DSperse patent** and formulated the **AI Safety Strategy** for the Canadian Ministry of AI.
- Deployed Local DeepSeek-R1-671B parameters on Mac Studio, quantized to 4-bit and Dynamic 1.5-bit pruning with 17 tokens/s throughput; Benchmarked against GCP-hosted equivalent using vLLM on a **Kubernetes** cluster.
- Experimented with the first verifiable distributed LLM on Qwen 2.5 leveraging **PEFT and LORA**, achieving near-zero computational overhead and automated testing via MCP, led to the creation of **DSperse**.
- Engineered scalable SR&ED RAG documentation pipeline utilizing Google Flan-T5, deployed on **VertexAI**.
- Implemented a Ranking Agent optimizing liquidity and slippage using DeepFM, and applying Direct Preference Optimization(DPO) and GRPO fine-tuning; containerized with **Docker** for scalable deployment.
- Deployed verified [soccer player detection](#) using external custom YOLOv model while preserving inference fidelity.

Machine Learning Developer

McMaster University

Dec 2022 – Dec 2024

Hamilton, Canada

- Developed an autonomous trading bot, with advanced risk mitigation strategies using Reinforcement Learning and FastDQN on TensorFlow; outperformed human traders by 185%.
- Designed and executed multi-objective network optimization for CSX Transportation Company using NetworkX and Gurobi; Scaled multi-threaded learning on HPC clusters (SLURM batch jobs).
- Engineered a novel **Hierarchical Reinforcement Learning framework**, improving conventional Multi-Armed Bandit solution by 13% with **Meta's PEARL**; a **Production-ready AI Agent** Library.
- Instructed ["Statistical Data Analysis"](#) course using Python at McMaster University ([Notebooks](#)).

Data Scientist

Nobitex

April 2022 – Dec 2022

Tehran, Iran

- Developed **scalable ensemble tree-based algorithms for customer clustering**, leveraging **PySpark** for distributed processing of 4+ million customers, doubling MAU from 4M to 8.5M within one year.
- Led cross-functional collaboration with the product team to design an **end-to-end marketing data modeling and ETL Pipeline** using Django and ORM Design.
- Defined over 50 financial KPIs, identifying company priorities and key metrics to support strategic growth.

- Developed blockchain-based fraud detection systems for OLTP wallet transactions and designed executive dashboards in Power BI.
- Utilized Web Engage for Customer Journey Mapping, achieving a 35% increase in conversion rate.
- Conducted market research and user survey analysis, driving a 10% increase in engagement.
- Performed sentiment analysis on Instagram and Twitter, extracting insights that improved retention rates by 30%.

Data Analyst

Oct 2020 – Dec 2021

Rocket Internet/SnappMarket

Remote, Canada

- Contributed to a \$100 million funded project for a **fully automated shopping experience** (Low-Cost Amazon Go), implementing indoor location tracking in collaboration with Rocket Internet.
- Developed an **optimized inventory reordering system** for 15 hypermarkets, automating algorithms for 8,000+ SKUs and coordinating supply chains across 200+ suppliers.
- Collaborated on **fraud detection**, retention, loyalty, and churn prediction programs through **A/B Testing**.
- Mentored two interns within the Data Team, now pursuing Math PhDs at UWaterloo and UMichigan.
- Delivered over 50 optimized SQL queries benefiting Marketing, Operations, and Finance teams.
- Established real-time automated email reports ensuring data accuracy for strategic decision-making.

EDUCATION

M.Sc. Computer Science

Dec 2022 - Dec 2024

McMaster University

Thesis: Multi-objective Weight optimization through **Hierarchical Reinforcement Learning in Decision Making**.

- Member of Degroote finance and investment council (DFIC Club) - Quantitative Finance Department.
- Completed the Vector Institute 2024 Deep Learning and **Reinforcement Learning** Summer School.
- Master Thesis Presented at the Canadian Operational Research Society CORS 2025 Conference.

B.Sc. Systems Engineering

Oct 2017 - Oct 2021

Sharif University

- Graduated with distinction from the Office of Exceptional Talents.

SKILLS

Core Expertise: Reinforcement Learning, Verifiable Inference, Distributed Inference, Quantization, Finetuning, AI Agents, MCP Server, Prompt Injection, LLM Evaluation, PEFT, LORA, DPO, GRPO, Statistical Testing

Programming: Python, Rust, R

Data Engineering: SQL (MySQL, PostgreSQL), MongoDB, Redis, Vector Databases

Cloud Platforms: GCP (Kubernetes, Vertex AI), AWS Certified (Lambda, EC2, SageMaker, Airflow, Elasticsearch)

Deep/RL Learning: PyTorch, TensorFlow, Hugging Face Transformer, ONNX Runtime, OpenAI Gym, Ray, Pearl, LangGraph, AutoGen

LLM Inference: TensorRT, vLLM, Mlx, KServe, llm-d, Dspy, llama.cpp

Data Tools/ DevOps: MLflow, Weights&Biases, PySpark, Docker, Kubernetes, Git, CI/CD pipelines, GraphQL, Bash

PUBLICATION

TruthTensor: Evaluating LLMs through Human Imitation on Prediction Market under Drift and Holistic Reasoning(Publication) (Dataset)

JSTprove: Pioneering Verifiable AI for a Trustless Future (Publication) (Sourcecode)

DSperse:A Framework for Targeted Verification in Zero-Knowledge Machine Learning (Paper) (Sourcecode).

HONORS AND AWARDS

CFA Institute Research Assistantship (\$10K)

Entrance Scholarship from Computational Science and Engineering (\$10K)

Excellence Scholarship Department of Mathematics at McMaster University(\$8K/year)

Merit-based Direct MBA-Finance Admission by the Office of Exceptional Talents at Sharif University

Member of Iran's National Elite Foundation (INEF)

Semifinalist in the National Mathematics Olympiad

National Youth Chess Champion